Case Report

ADRENAL INSUFFICIENCY SECONDARY TO BILATERAL ADRENAL HEMORRHAGE ASSOCIATED WITH ANTIPHOSPHOLIPID SYNDROME

Rashika Bansal, MD; Priti V. Nath, MD; Thanh D. Hoang, DO; Mohamed K.M. Shakir, MD

ABSTRACT

Objective: Antiphospholipid syndrome (APS) can involve multiple organ systems but endocrine manifestations are rare. In most cases adrenal insufficiency (AI) is the first endocrine manifestation of APS. The prompt diagnosis of AI is critical as this disorder is a life-threatening disease that may lead to fatal outcomes if left untreated. We present a case of AI associated with APS; the patient was diagnosed promptly and managed successfully.

Methods: The diagnosis of APS was based on a combination of clinical features (deep venous thrombosis and pulmonary embolism) and laboratory findings (lupus anticoagulant, anticardiolipin antibody, anti-beta-2 glycoprotein-I antibody), without alternative diagnosis to explain the clinical findings. AI was diagnosed by low morning serum cortisol with elevated adrenocorticotropic hormone (ACTH) level as well as an ACTH stimulation test.

Results: A 50-year-old male presented with deep venous thrombosis of the left extremity diagnosed by compressive ultrasound, and was subsequently diagnosed with a pulmonary embolism by computed tomography angiography and treated with heparin. Two days later, he developed hypotension and bilateral flank pain, and an abdominal computed tomography scan revealed bilateral adrenal hemorrhage. Laboratory results showed a serum cortisol of 3.3 mcg/dL (stress normal, 25 to 35 mcg/dL) and ACTH of 319 pg/mL (stress normal, 128 to 218 pg/mL), consistent with primary AI. Symptoms improved quickly with hydrocortisone therapy. The patient still required glucocorticoid therapy for at least 4 years thereafter.

Conclusion: In all cases of adrenal hemorrhage and infarction with unknown etiology, screening with lupus anticoagulant and anticardiolipin antibodies is imperative. Recognition of this high-mortality condition allows for appropriate screening and confirmatory tests leading to a prompt diagnosis and timely management. **(AACE Clinical Case Rep. 2020;6:e65-e69)**

Abbreviations:
ACTH = adrenocorticotropic hormone; **AI** = adrenal insufficiency; **APS** = antiphospholipid syndrome; **CT** = computed tomography; **SLE** = systemic lupus erythematosus

INTRODUCTION

Antiphospholipid syndrome (APS) is an autoimmune disease with acquired thrombophilia leading to recurrent thrombosis and pregnancy morbidity, and linked with a heterogenous group of antibodies known as antiphospholipid antibodies. APS can involve multiple organ systems but endocrine manifestations are rare. Adrenal glands are

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From the Division of Endocrinology, Department of Medicine, Walter Reed National Military Medical Center, Bethesda, Maryland.
Address correspondence to Dr. Thanh D. Hoang, Division of Endocrinology, Walter Reed National Military Medical Center, 8901 Wisconsin Avenue, Bethesda, MD 20889.
E-mail: thanh.d.hoang.mil@mail.mil.
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the most commonly involved endocrine glands in APS and in most cases adrenal insufficiency (AI) is the first endocrine manifestation of APS (1). Nonetheless, AI is one of the rarer complications of APS occurring in only 0.4% cases (2). Conversely, APS is diagnosed in less than 0.5% of all patients with AI (3). APS can cause hemorrhagic infarction of the adrenal glands and is typically associated with stress factors, including surgery, infection, or hemorrhagic diathesis associated with anticoagulation therapy (4).

The identification of this complication is extremely important as AI is a life-threatening disease that may lead to fatal outcomes if left untreated. A systematic review found the mortality to be 3.81% in patients with systemic lupus erythematosus (SLE)/APS complicated with AI, which is rather high given the young age of the patients (5). Despite having been reported in the literature previously, this clinical condition continues to baffle physicians due to its subtle presentation. We hereby describe a challenging case of AI associated with primary APS in a 50-year-old male presenting with acute abdominal pain and normal abdominal computed tomography (CT) initially. The patient subsequently developed systemic thromboembolic complications along with worsening abdominal symptoms leading to our final diagnosis of bilateral adrenal hemorrhage and primary AI during the course of his hospitalization.

METHODS

A literature search was conducted using the PubMed and Google Scholar databases for the diagnosis and management of adrenal insufficiency associated with APS. The diagnosis of APS was based on a combination of clinical features (deep venous thrombosis and pulmonary embolism) and laboratory findings (lupus anticoagulant, anticardiolipin antibody, anti-beta-2 glycoprotein-I antibody), without alternative diagnosis to explain the clinical findings. The diagnosis of AI was established by measuring 8:00 AM serum cortisol and plasma adrenocorticotropic hormone (ACTH), and a failure to achieve a peak serum cortisol level >18 to 20 mcg/dL after either 30 or 60 minutes following the administration of 250 mcg ACTH intravenously.

APS antibodies were measured by LabCorp, Burlington, NC. Serotonin release assay was developed by the LabCorp (Burlington, NC). Serum cortisol and ACTH were performed by the Laboratory at Walter Reed National Military Medical Center (Bethesda, MD). An axial CT scan was performed to visualize the adrenal glands.

CASE REPORT

A 50-year-old man was admitted with acute left lower extremity pain and edema. He was diagnosed with deep venous thrombosis of the distal left femoral vein extending to the popliteal vein by compressive whole leg ultrasound, and was started on rivaroxaban (Xarelto^{R}) but he developed a low-grade fever within 24 hours and pleuritic chest pain and dyspnea within 48 hours. A CT angiography confirmed that the patient had a pulmonary embolism (Fig. 1) and the patient was treated with an intravenous heparin drip. Two days following the heparin drip, the patient developed acute bilateral flank pain and hypotension; a CT scan of the abdomen showed 2 masses replacing the adrenal glands that were concerning for hematomas (Fig. 2).

Laboratory results showed a serum potassium of 4.9 mmol/L (normal, 3.7 to 5.2 mmol/L), a serum cortisol of 3.3 mcg/dL (nonstress normal, 7.2 to 19.4 mcg/dL; stress normal, 25 to 35 mcg/dL), ACTH level of 319 pg/mL (nonstress normal, 7 to 53 pg/mL; stress normal, 128 to 218 pg/mL), aldosterone of <1.0 ng/dL (normal, 0.0 to 3.0 ng/dL), and plasma renin activity of 7.17 ng/mL/hour (normal, supine, 0.15 to 3.95 ng/mL/hour). Serum antiphospholipid antibody testing showed a cardiolipin antibody (Ab) immunoglobin of 140 GPL/mL (normal, 0 to 14 GPL/mL), cardiolipin Ab immunoglobulin M (IgM) of 100 MPL/mL (normal, 0 to 12 MPL/mL) and cardiolipin Ab immunoglobulin A (IgA) >150 APL/mL (normal, 0 to 11 APL/mL). Further testing showed beta-2 glycoprotein 1 Ab immunoglobulin G of 103 GPI units (normal, 1 to 20 GPI units), IgM of 94 GPI units (normal, 0 to 32 GPI units), and IgA of 150 GPI units (normal, 0 to 25 GPI units). Although earlier testing was concerning for heparin-induced thrombocytopenia (based on a positive anti-PF4 enzyme-linked immunosorbent assay), confirmatory testing with a serotonin release assay was <1% (normal, 0 to 20%). The patient was treated with hydrocortisone 100 mg intravenously every 8 hours and the dose was subsequently tapered down. His hypoten-

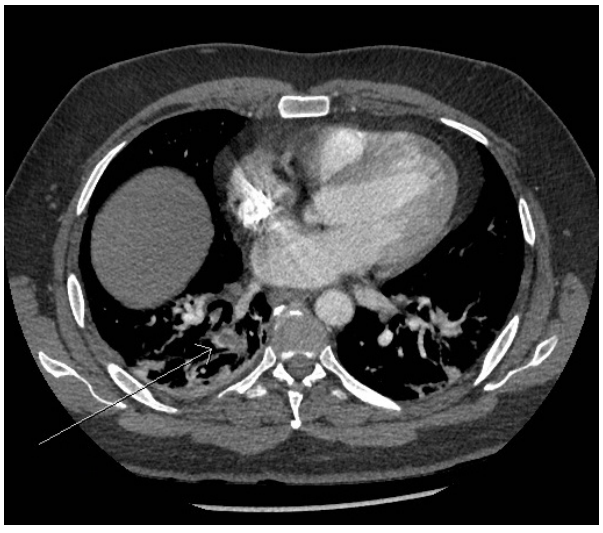


Fig. 1. Computed tomography angiography showing a pulmonary embolism.



Fig. 2. Abdominal computed tomography showing 2 adrenal masses concerning for hematomas.



Fig. 3. Follow-up adrenal computed tomography showing significant interval decrease in size of bilateral adrenal gland hematomas, favored to represent post-hemorrhage changes without evidence of underlying neoplasm.

sion dramatically improved after hydrocortisone and the abdominal pain resolved in 3 days. Upon discharge, he was placed on hydrocortisone 10 mg orally in the morning and 5 mg in the early afternoon with advice regarding sick day rules for increasing hydrocortisone dosing. Additionally, the patient was advised to continue warfarin therapy with monitoring of clotting times. At a 6-week follow-up visit, the patient was asymptomatic with normal blood pressure. Additional laboratory tests revealed normal plasma renin activity (3.1 ng/mL/hour) and aldosterone (12.3 ng/dL; normal, <28 ng/dL) levels and these values indicated that he had no profound hypoaldosteronism. Two years later an ACTH stimulation test showed a basal serum cortisol of 6.3 mcg/dL, a 30-minute serum cortisol of 7.4 mcg/dL, and a 60-minute serum cortisol of 7.7 mcg/dL. The basal plasma ACTH level was 230 pg/mL (normal, 5 to 50 pg/mL). Additionally, plasma renin activity and serum aldosterone levels (data not shown) remained normal, indicating no mineralocorticoid deficiency, although the ACTH stimulation test confirmed continued adrenal insufficiency.

The patient was doing well on hydrocortisone and warfarin treatment; he did not develop any adrenal crisis for the next 2 years. A follow-up ACTH stimulation test 2 years later also confirmed continued AI. An adrenal CT scan revealed a significant long-term interval decrease in the size of the adrenal hematomas with a hypo-attenuating focus in the right adrenal gland, favored to represent post-hemorrhage changes without convincing evidence of an underlying neoplasm, especially given the decrease in size compared to 4-years prior (Fig. 3). Presently, the patient is doing well on hydrocortisone and warfarin treatment.

DISCUSSION

Primary AI is caused by destruction of the adrenal cortex, leading to insufficient secretion of life-sustaining glucocorticoids and mineralocorticoids. The principal cause for primary AI in Western countries is believed to be autoimmune adrenalitis accounting for 70 to 90% of cases. Other causes are tuberculosis (7 to 20%), metastatic cancer, lymphoma or drugs (6). Only in 5% of cases the AI is due to unusual disorders, such as APS (3).

APS is a clinical entity characterized by thrombosis of both arterial and venous vasculature. It can occur as a primary condition or secondary to other systemic autoimmune diseases such as SLE. Diagnosis is based on a history of previous thrombotic episodes and the detection of antiphospholipid antibodies in serum on 2 or more occasions, at least 12 weeks apart (7). The spectrum of clinical manifestations of APS is vast and can potentially involve almost every organ system in the body. Endocrine manifestations of APS include AI, autoimmune thyroid disease, hypopituitarism, diabetes mellitus, and even thrombosis of the ovarian and testicular vascular beds (1).

The pathophysiologic mechanisms leading to AI in APS are not clearly established, however, proposed theories exist. The adrenal glands have a rich arterial supply but are drained by a single adrenal vein (3). Thrombosis of this limited venous supply predisposes to hemorrhagic infarction eventually causing hypoadrenalism (8,9). A novel mechanism has also been proposed involving cellular characteristics of zona fasciculata. The cells of this zone have a high density of late endosomes (organelles involved with

cholesterol exchange and protein sorting). The membranes of these organelles contain lysobiphosphatidic acid and antiphospholipid antibodies target these membranes. These antiphospholipid bodies react locally and cause accumulation of cholesterol within the cell, which leads to cell death and the release of lysosomal proteinases which in turn activates endothelial cells, favoring coagulation and causing micro-thrombosis (10). The fact that adrenal glands are anatomically symmetrical and have similar histologic structures help to explain the bilateral pattern associated with adrenal hemorrhage. Additionally, anticoagulation and heparin-induced thrombocytopenia could also be potential causes of bilateral adrenal hemorrhage.

The typical features of adrenal hemorrhage due to APS are usually nonspecific (weakness, fatigue, lethargy, altered mentation, etc.), although bilateral flank pain and hypotension should prompt a diagnosis of AI. Abdominal pain localized to lumbar, epigastric, umbilical, and the pericardial region was found in 66% of cases in a study published by Garcia et al (11). Fever (continuous or intermittent) was observed in 50% of cases, altered mental status, lethargy, and asthenia was seen in 12% of cases, whereas hypotension was seen in 19% of the cases (11). Espinosa et al (12) also published a case series of 86 patients in which 55% of patients had abdominal pain, 54% patients had hypotension, fever was seen in 40% of the patients, followed by nausea or vomiting (31%), weakness or fatigue (31%), and lethargy or altered mental status (19%) (12). Espinosa et al (12) also reported that out of 86 patients with APS syndrome and AI, 40% had deep venous thrombosis of the extremities, 17% had pulmonary embolism, and 19% had arterial occlusion (12). Our patient presented with symptoms of bilateral flank pain, hypotension, and tachycardia secondary to pulmonary embolism. The diagnosis of hypoadrenalism resulting from hemorrhage requires suggestive clinical symptoms, hormonal evaluation, and radiologic findings (11). A blunted response to 250 μg of ACTH with very high levels of baseline ACTH confirms primary AI, although a low serum cortisol with an elevated ACTH level is sufficient for a diagnosis of AI. However, the ACTH stimulation test was performed by the intensive care team. The presence of antibodies like lupus anticoagulant, anticardiolipin antibodies, and/or anti-β2-glycoprotein antibodies in medium to high titers without clinical evidence of concomitant rheumatologic conditions, such as systemic lupus erythematosus or SLE, further confirms APS (13). Adrenal hemorrhage is the principal finding (59%) seen in imaging of adrenal glands, while hemorrhagic infarction with vessel thrombosis is the most common finding (59%) in a histopathologic study (12). In the past, due to the insidious nature of presentation, a majority of cases of bilateral adrenal hemorrhage were diagnosed post-mortem. Abdominal CT scan remains the preferred modality to visualize the adrenal glands and can help in diagnosis at earlier stages. However, a CT scan in the beginning phase of the disease process may not reveal any abnormalities and should be repeated in cases where symptoms persist to look for morphologic changes (14). In our patient, this fact was reaffirmed and diagnosis was established only on repeat CT demonstrating bilateral adrenal hemorrhage. Once diagnosed with AI these patients need prompt treatment with stress doses of hydrocortisone initially and later adjusted to adequate replacement dose once stable. Mineralocorticoid replacement with fludrocortisone may also be started after aldosterone deficiency is confirmed. In our patient, testing revealed normal mineralocorticoid functions and hence he did not require fludrocortisone replacement. The management of thrombosis in patients with APS is a subject of controversy as it is a known cause of thrombophilia and at the same time lifelong anticoagulation increases the risk of bleeding. Although counterintuitive, anticoagulation is the mainstay of therapy in patients with APS and bilateral adrenal hemorrhage with the long-term benefit of prevention of future thrombotic events (15).

It is important that these patients follow-up for evaluation of their long-term adrenal function. Ramon et al (16) conducted a retrospective study of 16 patients with bilateral adrenal hemorrhage-adrenal infarction secondary to APS looking at long term prognosis of such patients. This study followed patients for a median of 3.5 years and demonstrated a relatively favorable prognosis among the survivors of the acute phase of adrenal involvement. It also suggested that adrenal dysfunction may partially recover in sporadic cases but is largely irreversible. To this end, measurement of early morning serum cortisol during follow-up is indicated to detect the patients that might have spontaneous recovery of the adrenal axis (16). We can conclude that our patient had chronic, irreversible primary AI without mineralocorticoid deficiency secondary to bilateral adrenal hemorrhage based on his initial CT scan, and on subsequent blood tests on follow-up where or when he failed ACTH stimulation tests 4 years later. Jahangir-Hekmat et al (17) reported the follow-up of 4 patients with acute bilateral adrenal hemorrhage and adrenal insufficiency for 6 to 19 years and found partial adrenal recovery in 3 and complete recovery in 1 patient. Ramon et al (16) conducted a retrospective study of patients with bilateral adrenal hemorrhage-adrenal infarction secondary to antiphospholipid syndrome between 1990 and 2010, and concluded that adrenal dysfunction is generally irreversible; however, partial recovery can occur (16).

CONCLUSION

There should be a high degree of suspicion for AI patients with APS who present with abdominal complaints, weakness, or asthenia and appropriate evaluation with blood pressure measurements, serum cortisol, ACTH, sodium, and potassium monitoring should be immediately performed. On the contrary, in patients with AI without

any known cause, the possibility of APS should be entertained after exclusion of common causes, irrespective of absence of thrombo-embolic manifestations. In all cases of adrenal hemorrhage and infarction with unknown etiology, screening with lupus anticoagulant and anticardiolipin antibodies is imperative. Awareness about this high mortality condition will allow for appropriate screening and confirmatory tests leading to a prompt diagnosis and timely management.

DISCLOSURE

The authors have no multiplicity of interest to disclose. The views expressed in this article are those of the authors and do not reflect the official policy or position of the Department of the Army, Department of the Navy, Department of Defense, or the United States Government.

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